Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

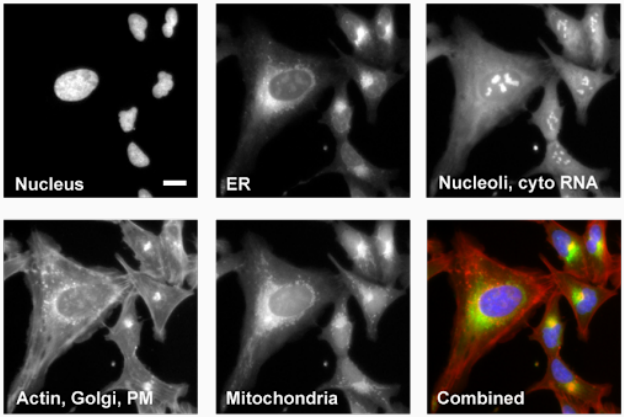
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University; Carpenter-Singh Lab, Broad Institute

10 November 2026

What is Cell Painting?

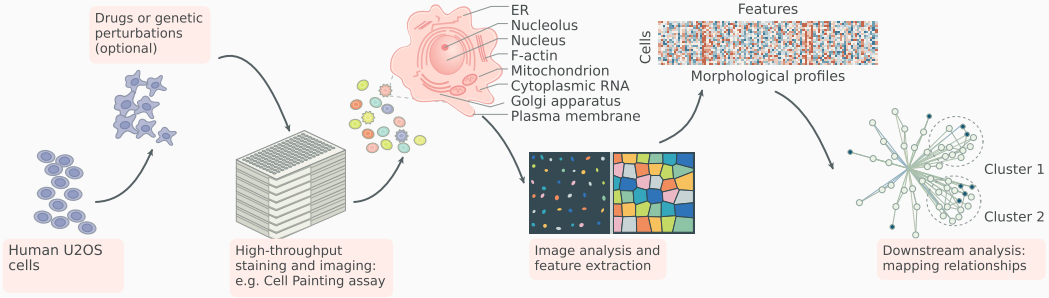
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

Was my compound tested? Nocodazole

A familiar biological question

Can morphology recover nocodazole's annotated mechanism?

- Nocodazole is present: `JCP2022_047835`.
- Annotated mechanism: tubulin polymerization inhibitor.
- Colchicine, vinblastine, vincristine and mebendazole are also present.
- For this example, use CellProfiler-derived profiles only.

Start with a simple query: "Find nocodazole and its annotations in JUMP."

JUMP_rr compounds; Repurposing annotations; checked 9 November 2026

Which compounds look similar to nocodazole?

Query: “Show nocodazole’s closest published matches.”

Annotated peer	Similarity	JUMP reagent ID
Colchicine	0.934	JCP2022_033814
CYT-997	0.916	JCP2022_056401
Parbendazole	0.916	JCP2022_110440
Oxibendazole	0.902	JCP2022_077096

- Four annotated peers occur among the five highest returned matches.
- Similar morphology suggests a useful biological connection.

CellProfiler-derived, batch-corrected profiles; published ranked subset, not an exhaustive screen.

JUMP_rr compound matches; checked 9 November 2026

Does nocodazole change cell morphology?

Published activity result

Corrected activity p-value: **0.00003**

- A detectable profile in this published analysis.
- Fewer neighboring cells: Cohen's $d = -3.306$.
- Lower nuclear form factor: Cohen's $d = -3.110$.

Follow-up question: could cell density contribute to the matches?

JUMP_rr compound features; checked 9 November 2026

Use JUMP to investigate a phenotype

Simple questions in JUMP_rr

1. Was nocodazole tested?
2. Does it have a detectable profile?
3. Which compounds look similar?
4. Which measurements change?

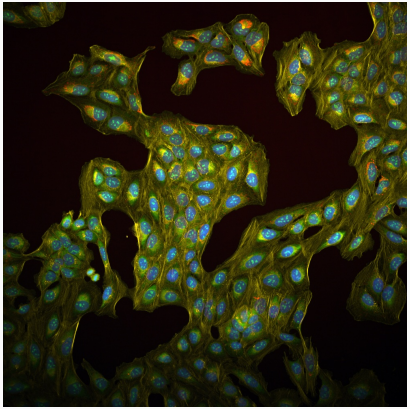
Then investigate the result

- Check images and matched controls.
- Check consistency across replicates.
- Test whether cell density explains part of the similarity.

One compound, one biological question, CellProfiler profiles.

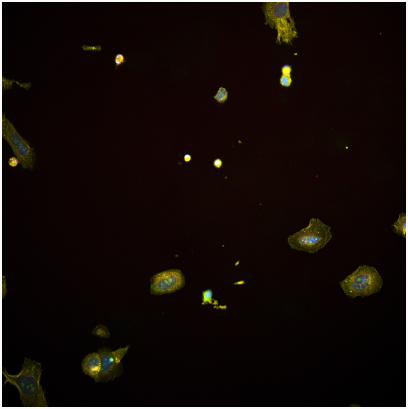
What do nocodazole-treated cells look like?

DMSO control



A02, site 1

Nocodazole



P15, site 1

Same source and plate: source_2 / 1086292020. Inspect density and cell appearance.

JUMP gallery / Ardigen previews

Live demo: investigate nocodazole's matches

From a simple query to an investigation

Investigate whether nocodazole retrieves other microtubule-disrupting compounds using CellProfiler profiles. Compare their retrieval against unrelated compounds, check consistency across replicates, and show representative images alongside the quantitative results.

My follow-up:

"Could this reflect cell density or toxicity? Check cell counts and quality-control measurements."

Finish with:

"Turn the result into one slide with an image comparison, a retrieval plot and the main limitation. Build a preview."

External agent and backends on Oppy; forwarded interface on the MacBook.

What we know, and what we will check

Checked before the demo

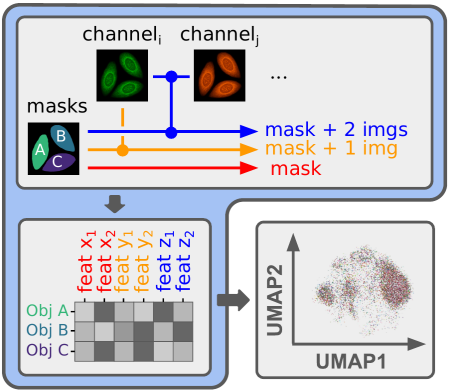
- Nocodazole and 19 annotated peers are present.
- Colchicine is its top returned match.
- Images and same-plate DMSO controls are available.

The live investigation

- How consistent are the replicates?
- Are annotated peers preferentially retrieved?
- How much is explained by density or QC?

From a recognizable phenotype to a testable biological hypothesis.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, Vendrell Lab, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

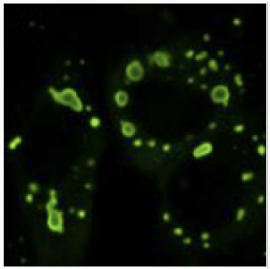
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

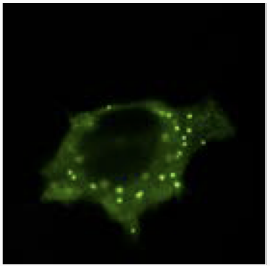
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Unpublished data (preprint coming soon)

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

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Runxi Shen

JUMP partners

JUMP supporting partners

JUMP collaborators

Scientific:
Horizon Discovery,
a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

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References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

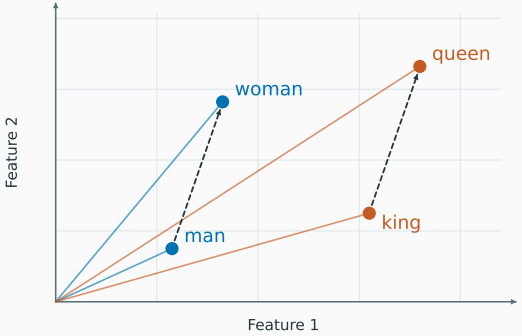
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

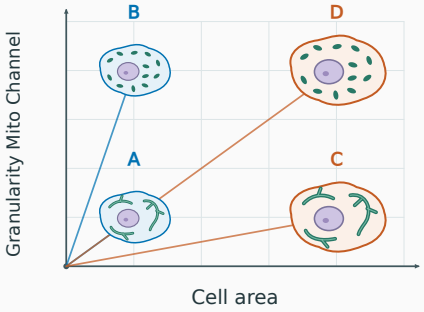
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

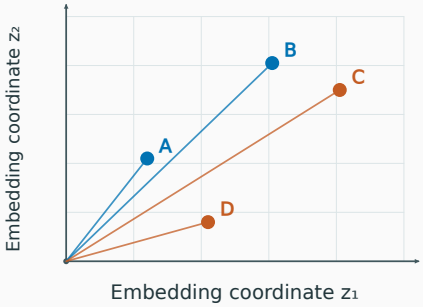


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



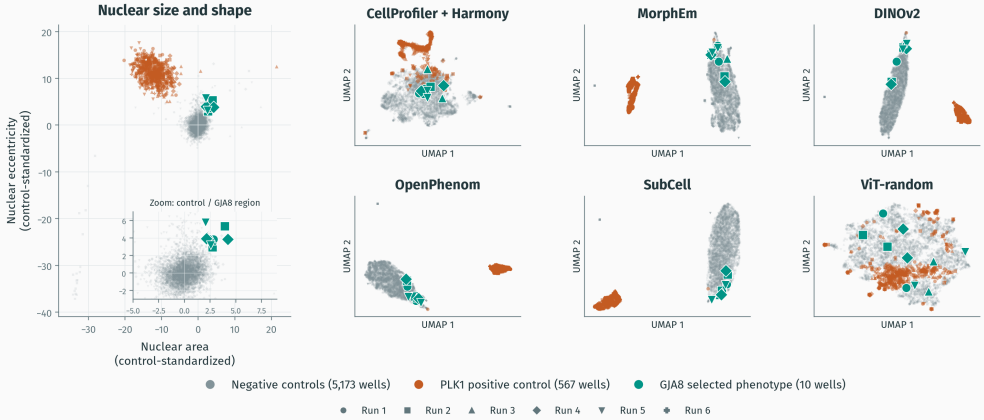
Control

Perturbation

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

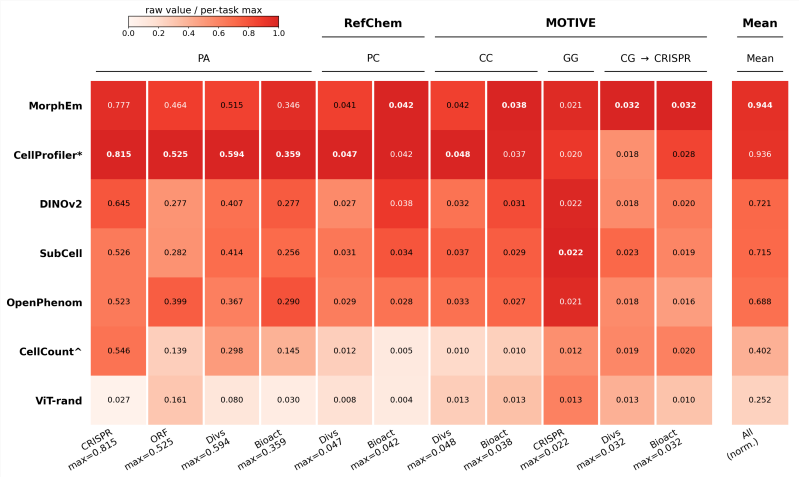
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

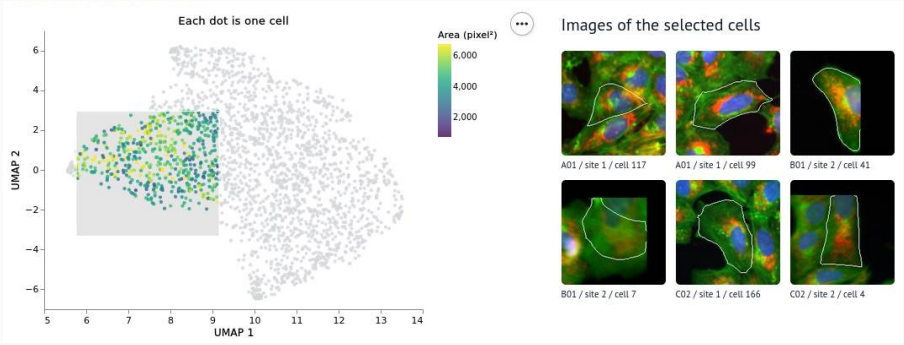


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

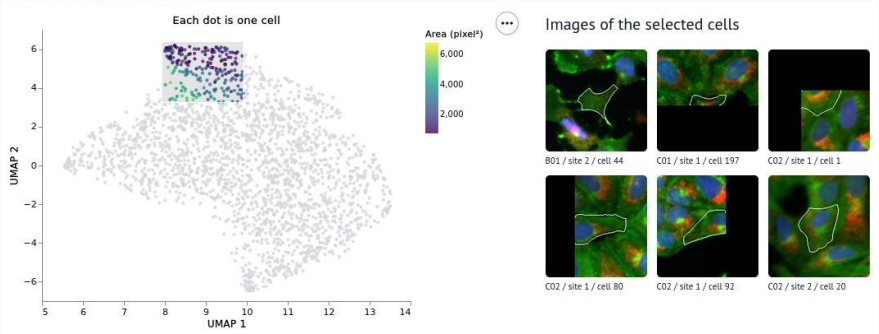
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

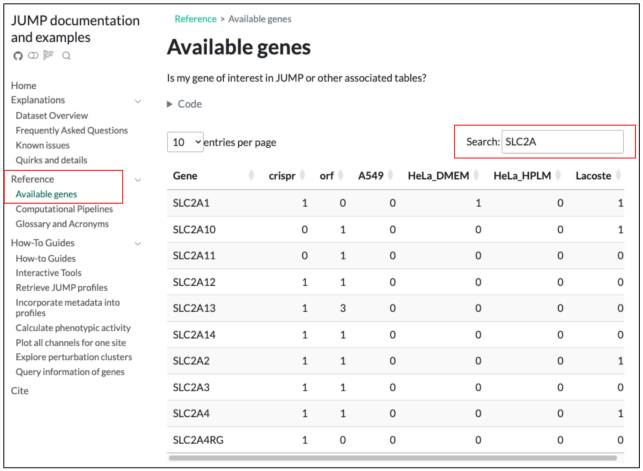
Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the nocodazole query; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording

1. Is my gene represented?

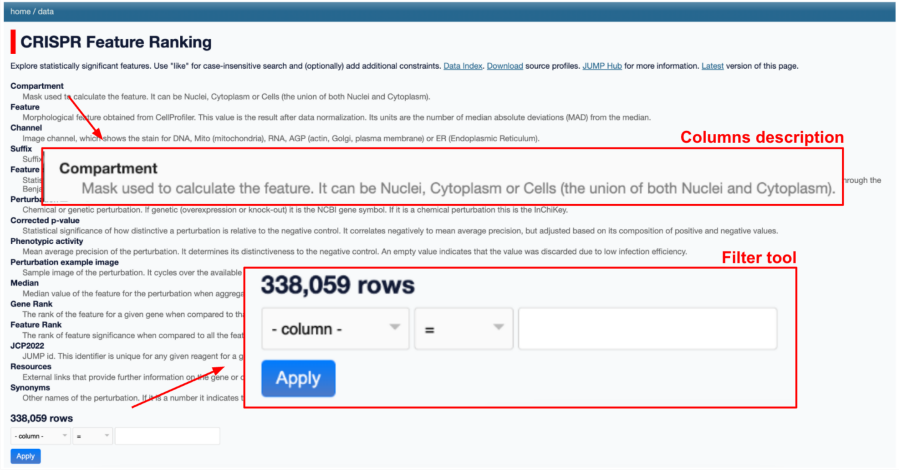


Gene	crispr	orf	A549	HeLa_DMEM	HeLa_HPLM	Lacoste
SLC2A1	1	0	0	1	0	1
SLC2A10	0	1	0	0	0	1
SLC2A11	0	1	0	0	0	0
SLC2A12	1	1	0	0	0	0
SLC2A13	1	3	0	0	0	0
SLC2A14	1	1	0	0	0	0
SLC2A2	1	1	0	0	0	1
SLC2A3	1	1	0	0	0	0
SLC2A4	1	1	0	0	0	1
SLC2A4RG	1	0	0	0	0	0

Read it as: coverage by dataset and perturbation type, not evidence of a phenotype.

Broad Institute, JUMP Hub tutorial, Fig. 3; CC BY 4.0. Original annotations retained.

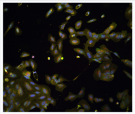
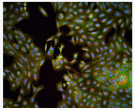
2. Read the column definitions first



Look at the red boxes: column definitions explain what the numbers mean; filters define the question.

Broad Institute, JUMP Hub tutorial, Fig. 1; CC BY 4.0. Original annotations retained.

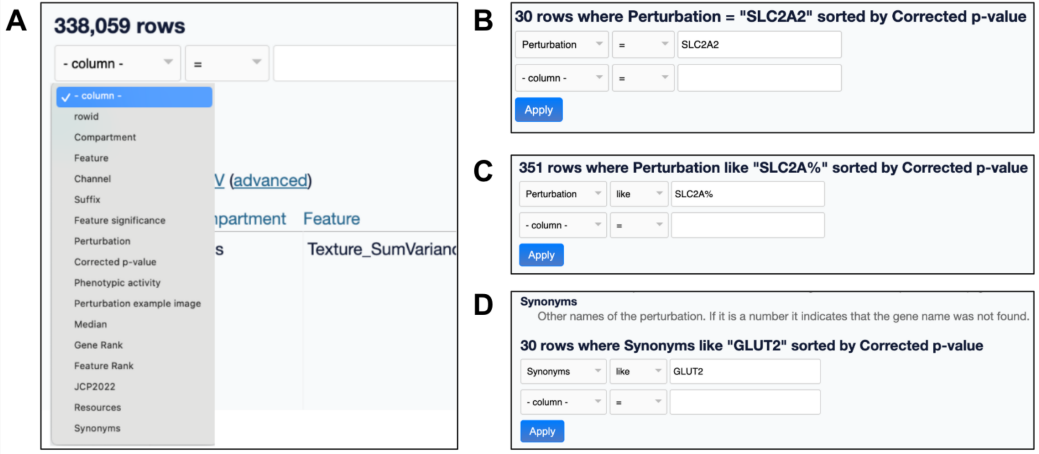
3. What does a result row represent?

Compartment	Feature	Channel ▲	Suffix	Feature significance	Perturbation	Corrected p-value	Phenotypic activity	Perturbation example image	Median	Gene Rank	Feature Rank
Cells	Intensity_MassDisplacement	RNA		1e-05	SNRNP200	0.00018	1.0		-0.965	999999	27
Nuclei	Texture_AngularSecondMoment	RNA	_5_03_256	0.0	CECR1	0.00018	0.77241		-0.203	999999	22

Read it as: a feature–perturbation summary, not one cell or an independent replicate.

Broad Institute, JUMP Hub tutorial, Fig. 2; CC BY 4.0. Published screenshot unchanged.

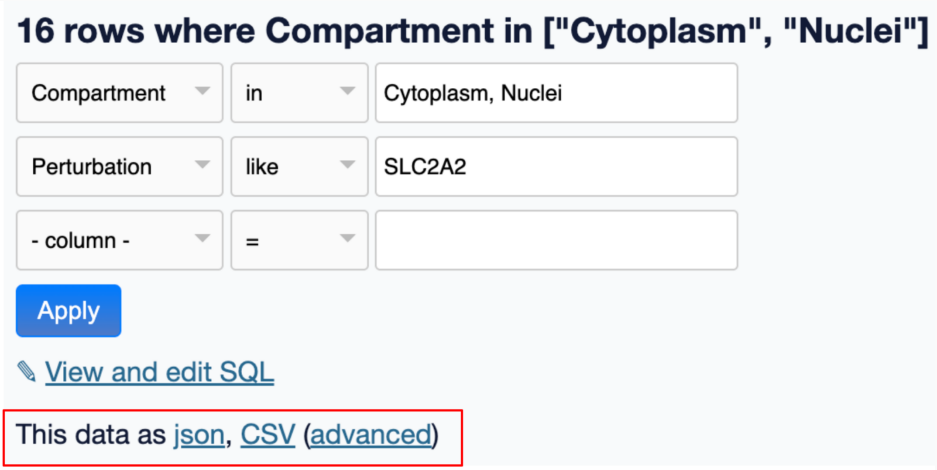
4. Ask a precise question with filters



Compare the panels: exact symbol, family pattern and synonym searches can return different sets.

Broad Institute, JUMP Hub tutorial, Fig. 4; CC BY 4.0. Original panels retained.

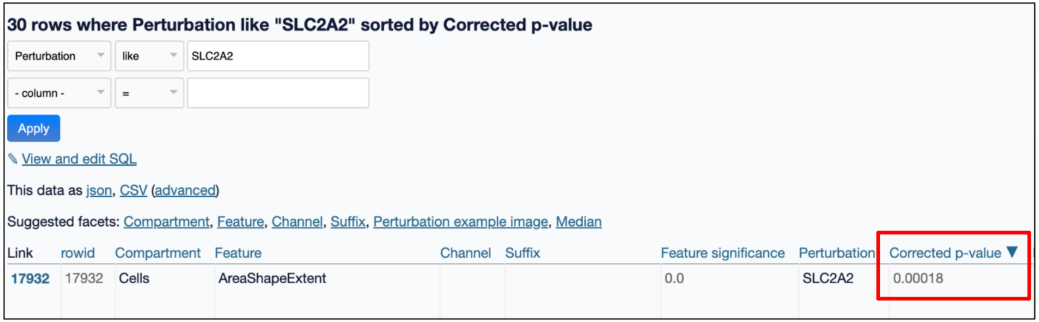
5. Narrow the question and save the result



Read it as: a filtered view of a ranked subset, not the full screen. Save the filters and source.

Broad Institute, JUMP Hub tutorial, Fig. 5; CC BY 4.0. Original annotations retained.

6. Is the perturbation phenotypically active?



30 rows where Perturbation like "SLC2A2" sorted by Corrected p-value

Perturbation | like | SLC2A2

- column - | = |

Apply

View and edit SQL

This data as json, CSV (advanced)

Suggested facets: Compartment, Feature, Channel, Suffix, Perturbation example image, Median

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation	Corrected p-value ▼
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2	0.00018

Read the boxed corrected p-value: activity is tested per perturbation, not separately for every row.

Broad Institute, JUMP Hub tutorial, Fig. 6; CC BY 4.0. Historical SLC2A2 example.

7. Which morphological features change?

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2
21804	21804	Nuclei	NeighborsSecondClosestDistance		_1	0.0	SLC2A2
22777	22777	Cells	NeighborsPercentTouching		_Adjacent	0.0	SLC2A2
33343	33343	Cytoplasm	RadialDistributionFracAtD		_mito_tubeness_12of20	0.0	SLC2A2

Read feature significance separately. A displayed 0.0 is rounded; also inspect rank and effect size.

Broad Institute, JUMP Hub tutorial, Fig. 7; CC BY 4.0. Historical SLC2A2 CRISPR example.

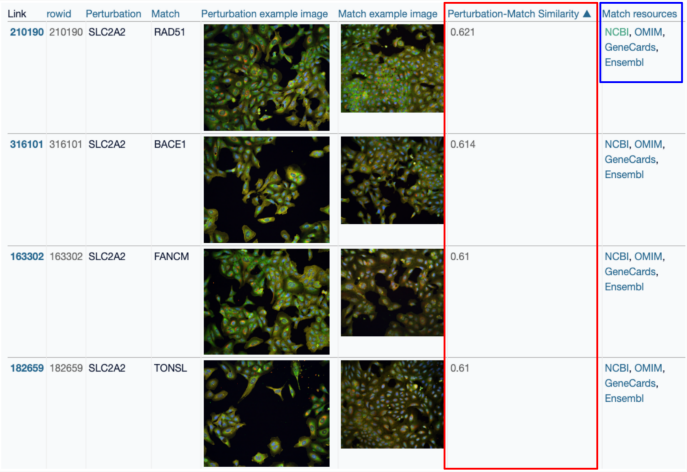
8. Keep overall activity and features distinct

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance ▼
394755	394755	Cytoplasm	AreaShapeMinorAxisLength			8e-05
293369	293369	Cells	AreaShapeMinorAxisLength			0.00011
126201	126201	Cells	AreaShapeMinFeretDiameter			0.00036
292792	292792	Cytoplasm	AreaShapeMinFeretDiameter			0.00036

ORF example: significant individual features need not imply significant overall activity.

Broad Institute, JUMP Hub tutorial, Fig. 8; CC BY 4.0. Historical SLC2A2 ORF example.

9. What does a profile match tell us?

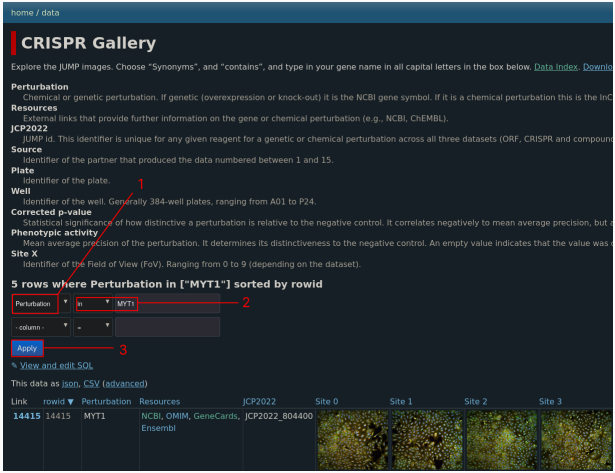


Link	rowid	Perturbation	Match	Perturbation example image	Match example image	Perturbation-Match Similarity ▲	Match resources
210190	210190	SLC2A2	RAD51			0.621	NCBI, OMIM, GeneCards, Ensembl
316101	316101	SLC2A2	BACE1			0.614	NCBI, OMIM, GeneCards, Ensembl
163302	163302	SLC2A2	FANCM			0.61	NCBI, OMIM, GeneCards, Ensembl
182659	182659	SLC2A2	TONSL			0.61	NCBI, OMIM, GeneCards, Ensembl

Read the signed similarity and follow the links. A match suggests follow-up, not a shared mechanism.

Broad Institute, JUMP Hub tutorial, Fig. 9; CC BY 4.0. Historical SLC2A2 example.

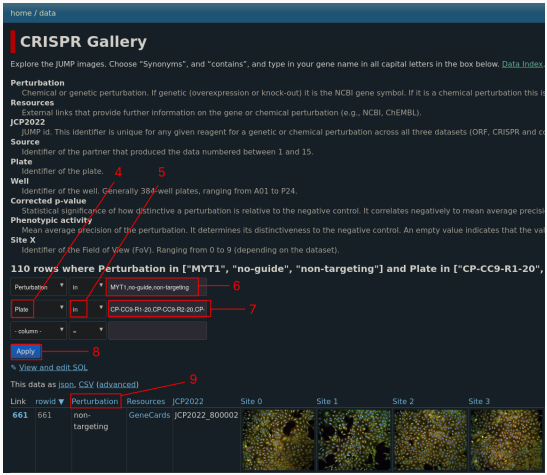
10. Find the perturbed wells and images



Follow steps 1–3: select the perturbation, apply the filter, and identify its source and plates.

Broad Institute, JUMP Hub gallery tutorial, steps 1–3; CC BY 4.0. Historical MYT1 example.

11. Compare with appropriate negative controls



Follow steps 4–9: match source and plate; check cell density, staining and focus before interpreting biology.

Broad Institute, JUMP Hub gallery tutorial, steps 4–9; CC BY 4.0. Genetic controls shown; compounds use DMSO.